

Rene Santillana, EIT

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EDUCATION

B.S. in Mechanical Engineering, Emphasis in Nuclear Engineering

Kansas State University • 3.92/4.0 GPA (Magna Cum Laude)

May 2021, Manhattan, KS

EXPERIENCE

Mechanical Engineer – Ground & Space Teams

Added Valued Solutions (AVS) | US

October 2024 – Present, Lansing, NY

- Support research, development, and deployment of ground and spaceflight hardware for government, private, and aerospace customers, including concept development, prototyping, testing, and system integration
- Develop detailed mechanical designs and prototypes to support technical proposals and proof of concept demonstrations
- Perform structural, thermal, and environmental analysis to ensure compliance with missions, testing, and operations
- Plan and support environmental testing activities, including random vibration and thermal vacuum testing. Contribute to test plans, procedures, and reports
- Directly interface with customers and end users during technical meetings and design reviews to gather requirements, present progress, and incorporate feedback
- Coordinate with vendors and suppliers to ensure equipment meets performance specifications and project requirements
- Contribute to system documentation, including technical descriptions, specifications, and internal engineering reports

Mechanical Design Engineer

Salina Vortex Corporation

June 2022 – October 2024, Salina, KS (Remote/Hybrid)

- Led new product development strategy, overseeing construction estimation, design, testing, building, and production processes from conception to completion, while serving as the main interface
- Designed and evaluated mechanical components under complex loading conditions using analytical and numerical methods
- Communicated with vendors and managers to reduce purchase and manufacturing costs for projects
- Executed detailed calculations to determine actuation forces and structural margins
- Coordinated cross-functional efforts across design, manufacturing, and sales to support continuous improvement initiatives
- Mentored interns in CAD modeling, engineering standards and documentation practices
- Designed and monitored fabrication of over 500 parts and assemblies
- Reviewed over 5000 engineering drawings to ensure technical compliance with ANSI, AWS, & ISO standards

Graduate / Undergraduate Research Assistant

Kansas State University, Mechanical & Nuclear Engineering Department

Aug 2020 – March 2022, Manhattan, KS

NASA Lunar Simulation:

- Standardized methods for creating and applying albedo response functions
- Utilized active shielding simulation data to compute incident lunar surface flux and direct crew-volume radiation flux for lunar scenarios
- Developed features of C++-based particle propagation simulation software (ASPP)

NASA Active Radiation Shielding Simulation:

- Analyzed and quality checked over 2000 active shielding simulations outputs
- Develop code to facilitate post-processing of erroneous simulations results
- Drafted monthly reports for NASA sponsors and principal investigators

PROJECTS

Cornell High Energy Synchrotron Source (CHESS) Sector 6 Front End

- Conducted thermal FEA to ensure safe dissipation of high beam power through actively water-cooled components

- Designed UHV-compatible mechanical components and support frame structures for installation in highly constrained beamline environment
- Supported design documentation and analysis used for review and integration into the synchrotron facility

CHES Custom Micro Ion Chamber

- Modified standard AVS Micro Ion chamber to enhance detection performance for soft X-rays by allowing the particles to enter the gas volume with minimal energy loss and scattering

University of North Dakota - Rendezvous & Operations for Autonomous Docking and Servicing Satellite Program (UND ROADS)

- Redesigned shock resistant docking structure, doubling effective magnetic docking range
- Planned, oversaw, and conducted random vibration analysis and testing for two flight-unit satellites
- Conducted thermal vacuum testing of two flight-unit satellites
- Supported assembly of the two flight-unit satellites

WashU Lunar-Libs Analyses and Detection (2LEAD) – Ongoing:

- Designing precision XY scanning mirror and linear stage mechanism capable of microradian and micron level resolution to be used in a space environment
- Leading random vibrations analysis efforts across prototype, protoflight, and acceptance test levels
- Developing non-pressurized enclosure for integration with a lunar rover
- Designing structural and thermal solutions to enable heat dissipation and retention in lunar environment

L3Harris Project

- Served as team leader for a multinational AVS team during a 10-day onsite customer repair and support effort

Research on the optimization of molten salt reactors – Idaho National Laboratory

- Focused on developing a model to determine the theoretical products of the MSR reaction cycle

CERTIFICATIONS

Engineer-In-Training (EIT) Certification – Kansas, March 2024

SolidWorks Certification – Associate, May 2019

SKILLS

- Programs: MCNP, Inventor, SolidWorks, Mecway, DraftSight, Infor CloudSuite, GitHub
- Languages: Python, C++ Python, C, MATLAB, LaTeX, Bash
- Foreign Languages: Spanish; Bilingual
- Other: Nuclear lab equipment

PUBLICATIONS

Design of high-power load front ends for two upcoming beamlines at CHES

Poster presentation, MEDSI2025 – 13th International Conference on Mechanical Engineering Design of Synchrotron Radiation Equipment and Instrumentation, Lund, Sweden, September 2025

Systematic modeling of electrostatic radiation shields for deep space flight

Published in Radiation Physics and Chemistry, Volume 193, April 2022, 110007

Space radiation electrostatic shielding scaling laws: Beam-like and isotropic angular distributions

Published in Journal of Applied Physics, Volume 130, January 2022, 034903